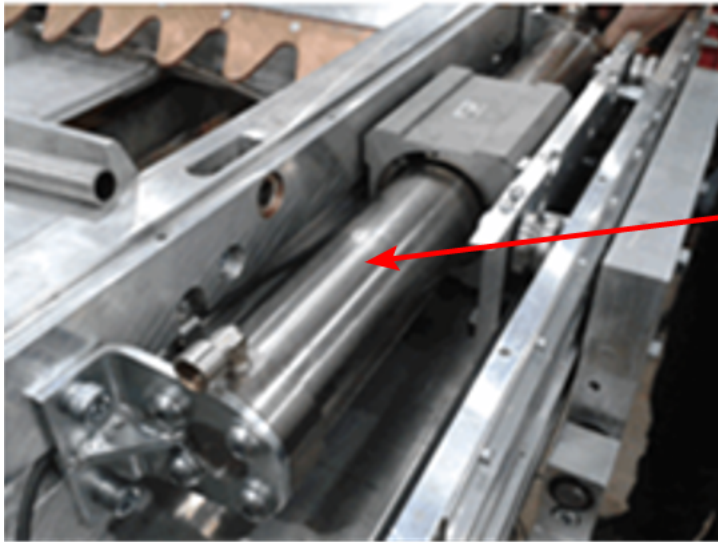




Note: There is a requirement for a clip to hold the cable in here

Figure 9.11-34 Example of a technician checking that the cylinder movement is correct.

12. Lower the guide plate between the cylinder and the carriage plate and secure with slotted button-head screws on top

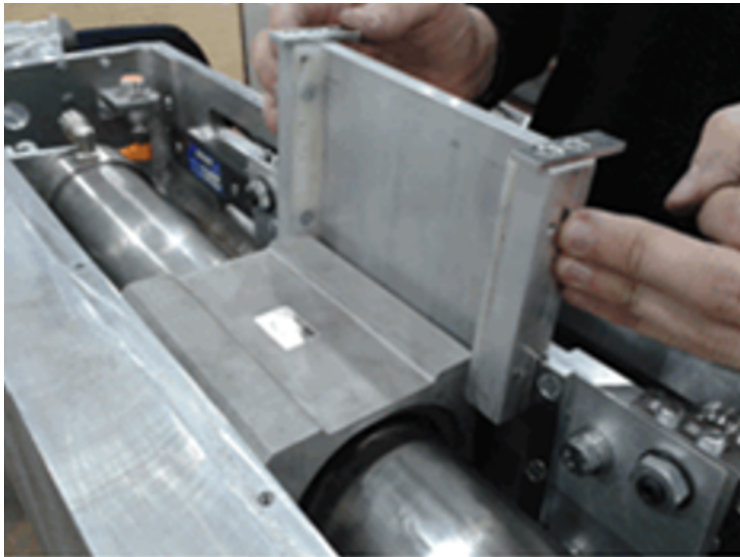


Figure 9.11-35 Example of a technician lowering the guide plate into place.

13. It is advised to use steel shims behind the cylinder mounting brackets to maintain a gap of 0.5–1 mm between cylinder and guide plate.

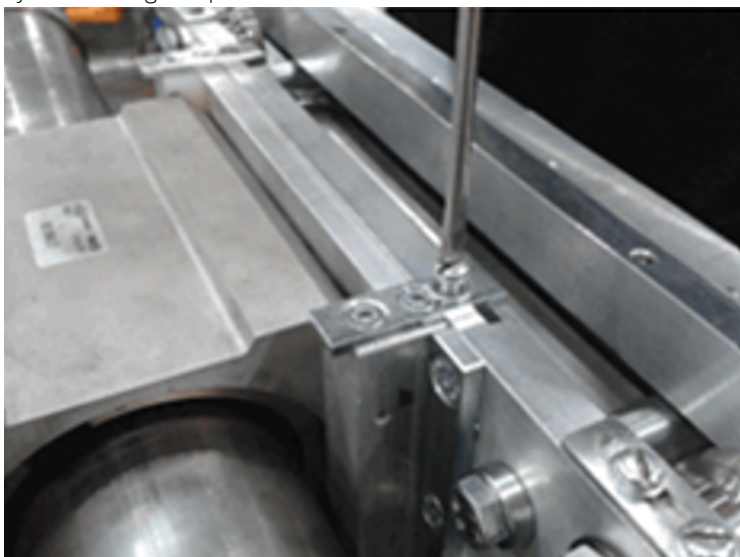


Figure 9.11-36 Example of a technician screwing down the locating button-head screws. Note the 3 slots to allow shutter alignment adjustment.



Figure 9.11-37 The outboard shutter guide plate assembly.

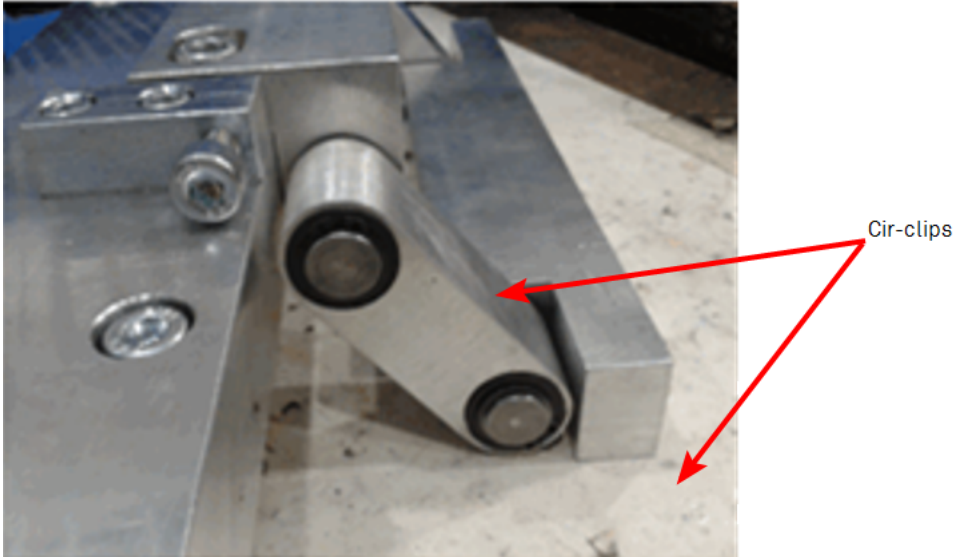


Figure 9.11-38 The pivot arrangement to allow shutter expansion



Figure 9.11-39 Example showing how to mount the outboard guide plate to align with the inboard side guide plate.

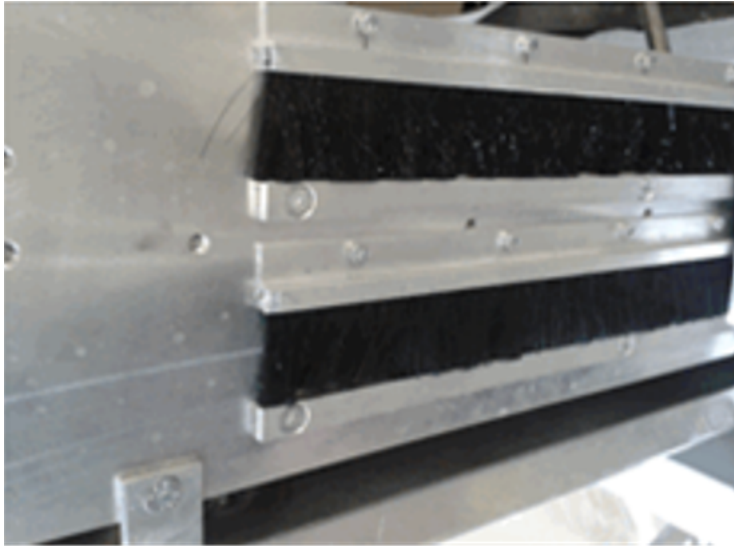


Figure 9.11-40 The brush strips used to keep debris out of the enclosure.

Testing for Air Leaks on the Assembly/System

Small air leaks may have an effect on the performance of the machine, in regards to vacuum.

A “Gross” leak test can be carried out using a suitable product as Swagelok “Snoop” leak detector

http://www.swagelok.com/search/find_products_home.aspx?show_results=Y&item=5e208092-ed6c-4251-9202-ed8a2aae5811

If a helium leak detector with a “sniffer” adaptor is available, this could be used to carry out a “fine” test, which is more accurate at lower leakage rates.

9.12 Coating Shield Rotational Movement

The coating shield that is located between the evaporation source and the process drum can be rotated downwards for cleaning and set-up purposes.

An internal cross shaft is provided which is used as a pivot point.

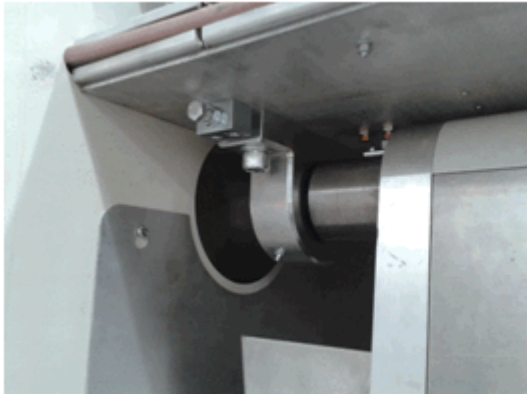
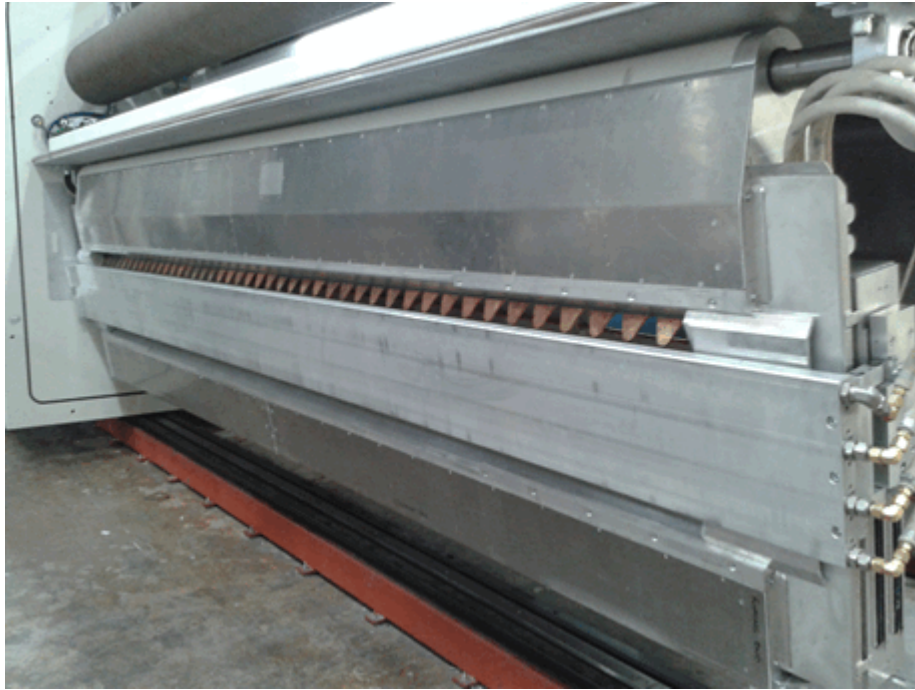


Figure 9.12-1 The coating shield and its pivot points

The coating shield assembly is manufactured off machine and is offered up to the machine using a fork truck.

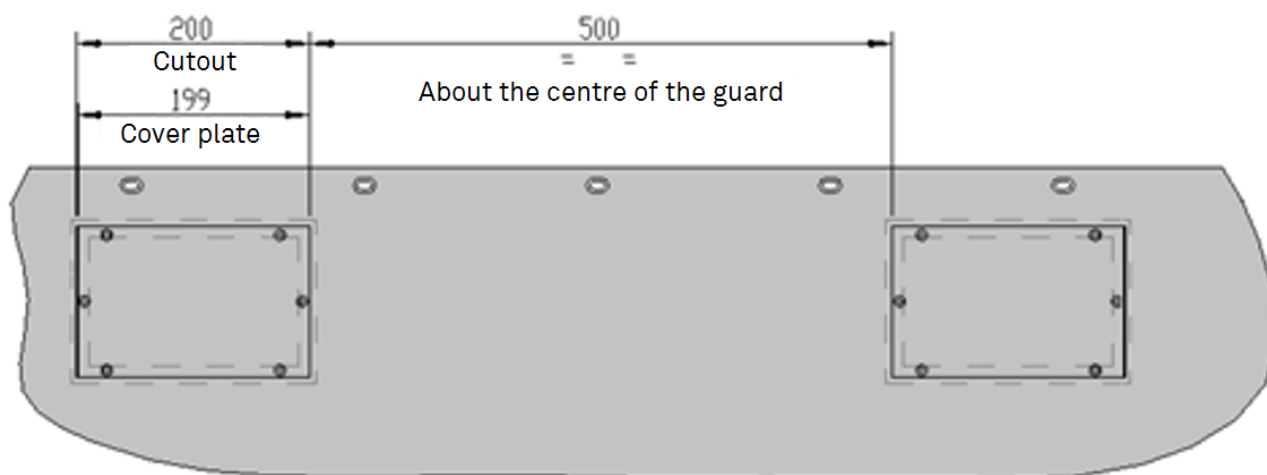


Figure 9.12-2 The cut outs in the coating shield

There are cut outs with blanking plates in the shield to allow the lifting and installation by the fork truck.

Removal of Shield Assembly

Reference should be made to mechanical drawings 417G155/1 and 417G587/2.

1. The shield should be lowered to its vertical position.
2. Remove the lifting steel cables by removing the circlips (B29) either side of the lifting pin (M46) and removing the pin.(417G587/2.)
3. Remove the air supply from the machine and lock it out.
4. Remove the air connections
5. Remove the electrical leadthrough that connects the position sensors located in the shield and loosen the leadthrough between the zone plates.



Figure 9.12-3 The lifting cable installed in the shield

6. Remove plates in shield as shown in figure 11.12-2
7. Take the weight of the assembly on a fork truck with the forks. Place a piece of wood on each fork to stop potential sliding

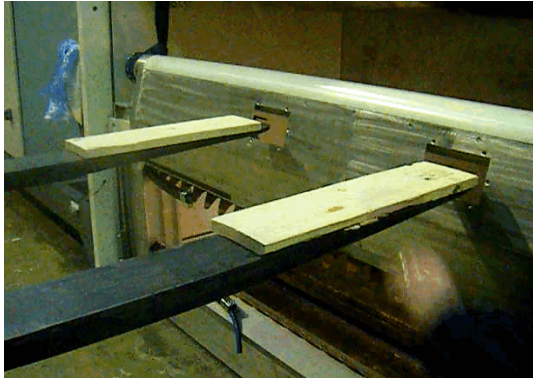


Figure 9.12-4 Example of the forks on a fork lift truck being placed into the coating shield

8. Once supported, loosen off bolts securing bearing housings (B31) to the winding mechanism.
9. Lower the shield away from the winding mechanism and take away.

Replacement of Wire Ropes

Reference should be made to mechanical drawing 417G587/2.

The replacement steel ropes should be purchased from Bobst Manchester Ltd, as the details of rating, etc. are known by our supplier. Also the requirements in regards of the size of the pre-installed clamp are known.

Take the following steps to remove and replace the wire ropes:

1. Lower the shield to the vertical position, so no tension is applied to the ropes.
2. Remove the lifting steel cables by removing the circlips (B29) either side of the lifting pin (M46) and removing the pin.(417G587/2)
3. Loosen the screws from the clamp (M04) and feed the cable through the slot in the drum wall (M02) and pull the rope clear.
4. Carry this action out on the other side.

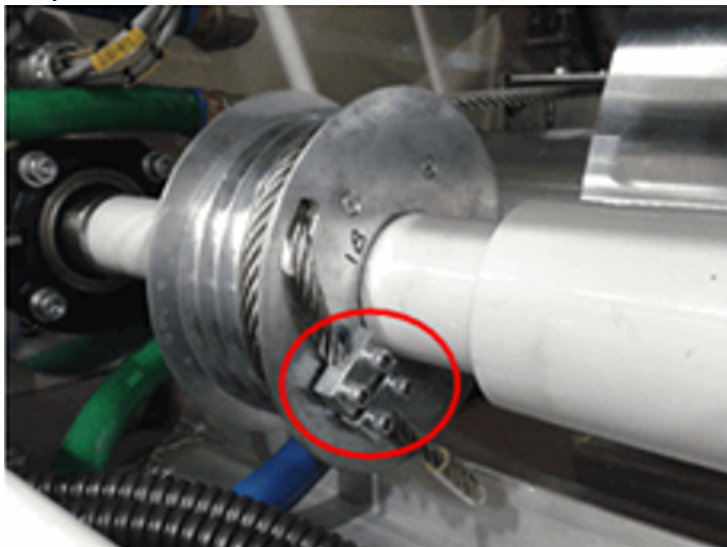


Figure 9.12-5 Example of the lifting cable before starting removal.

5. Loosen the 4 cap head bolts slightly, and loosen the jacking bolt positioning hexagonal nuts to allow the bracket to slide to its lowest position.

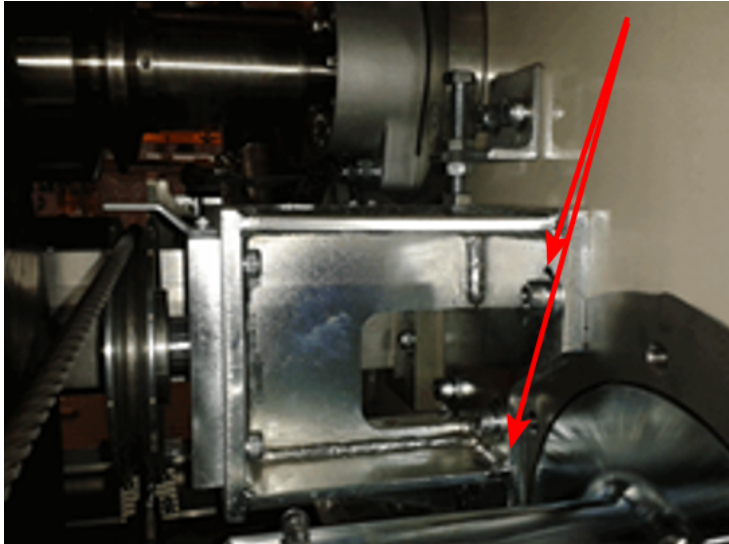


Figure 9.12-6 The sheave support bracket

6. Install the end of the lifting cables ensuring the lifting pin (M46) is secured with circlips (B29) either side.
7. To ease the pin through, use a small amount of vacuum grease.



Figure 9.12-7 The lifting cable installed in the shield

Feed the cable over the sheave, into the second groove of the cable drum (M02) and wrap approximately 1.5 turns around the cable drum (M02)

8. Carry this action out on the other side.

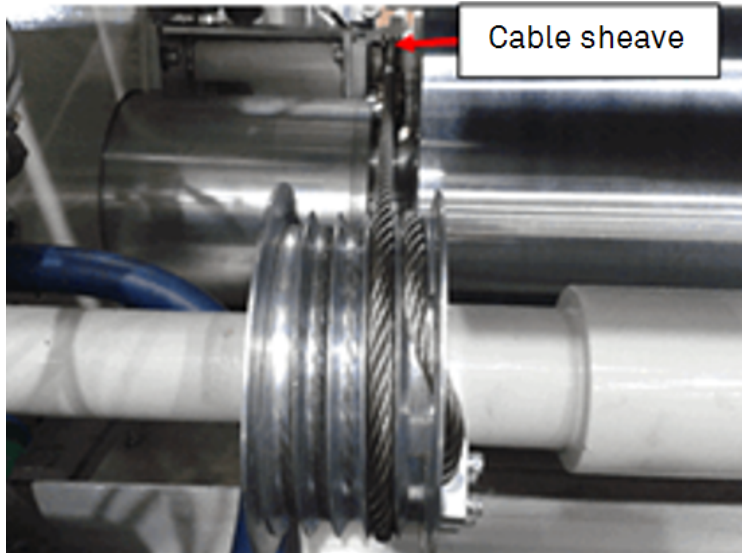


Figure 9.12-8 The wire rope passed over the winding drum

9. Feed the cable through the slot in the drum wall (M02) and apply a small amount of tension to the cable by hand then secure with the clamp (M04). Tighten screws to 10Nm.
10. Carry this action out on the other side.

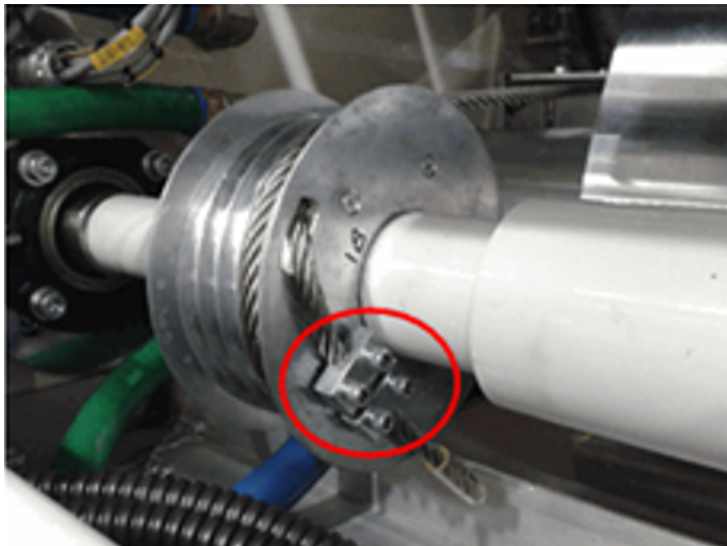


Figure 9.12-9 Example of a clamped cable.

Raise the shield carefully ensuring no clashes with other equipment, particularly the drum or the 3 positioning sensors.



Figure 9.12-10 Example of the coating shield lifted up into location.

If one end of the shield is lower and needs to be levelled, when raised by the cables, due to the tension being different, additional adjustment can be made using the cable adjustment bracket shown.

Note it is easier to make the adjustment when the shield is fully open, and then close it again to re-check to gap.

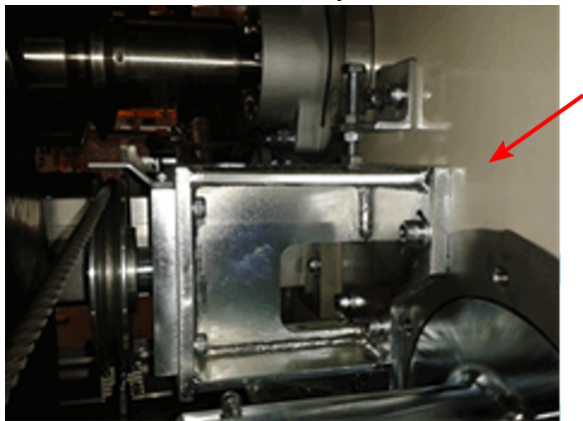


Figure 9.12-11 The sheave support bracket, jacking bolt



Note !

Adjust the height using the two nuts. (The hexagonal bolt is not to be turned)

Installation and Alignment after Removal of Shield

Reference should be made to mechanical drawings 417G155/1 and 417G587/2.

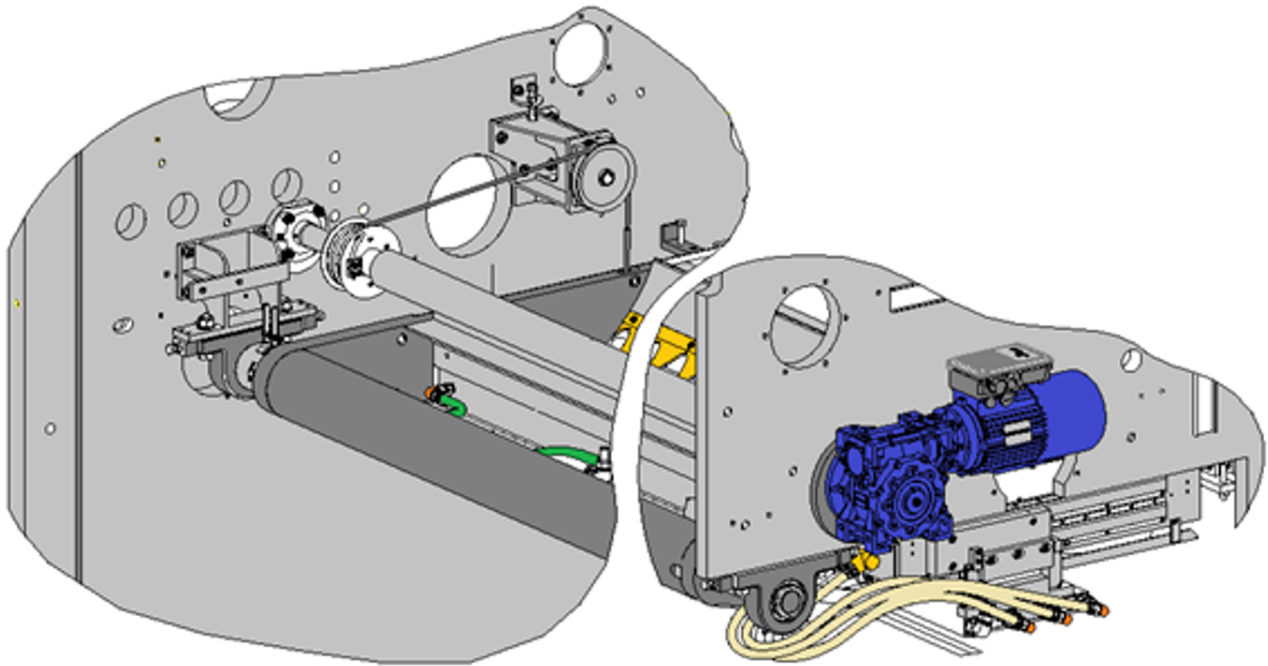


Figure 9.12-12 Shield winch system

This procedure can be used for replacing the shield into the machine, or replacing the steel ropes and assumes that the motor drive is operational.

1. Raise the complete shield assembly into position using a fork truck or similar device and secure the bearing housings (B31) at both ends onto the machine, with the bolts supplied
2. Set the bearing housings to the middle position of their adjusting slots, for horizontal adjustment.

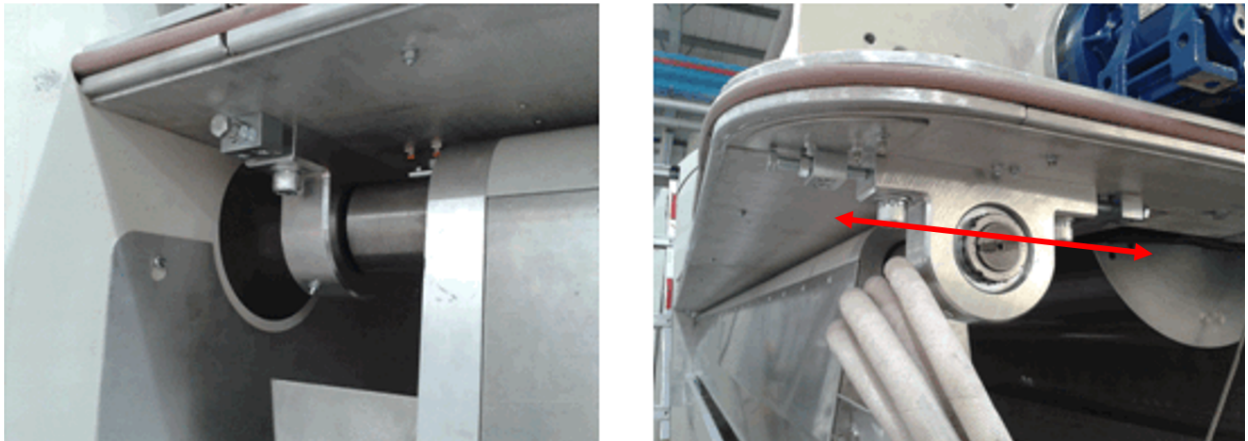


Figure 9.12-13 The pivot bearing housings in location, with positional jacking screws located either side.

	Note !
	The bracket on the left side, where the bolts pass through allows for movement.

3. To allow the shield to be raised up to the process, drum, it needs to be positioned so that the distance from the face-plate to inner edge of the inboard arm to 273mm. Use two pieces of wood to move the shield as necessary

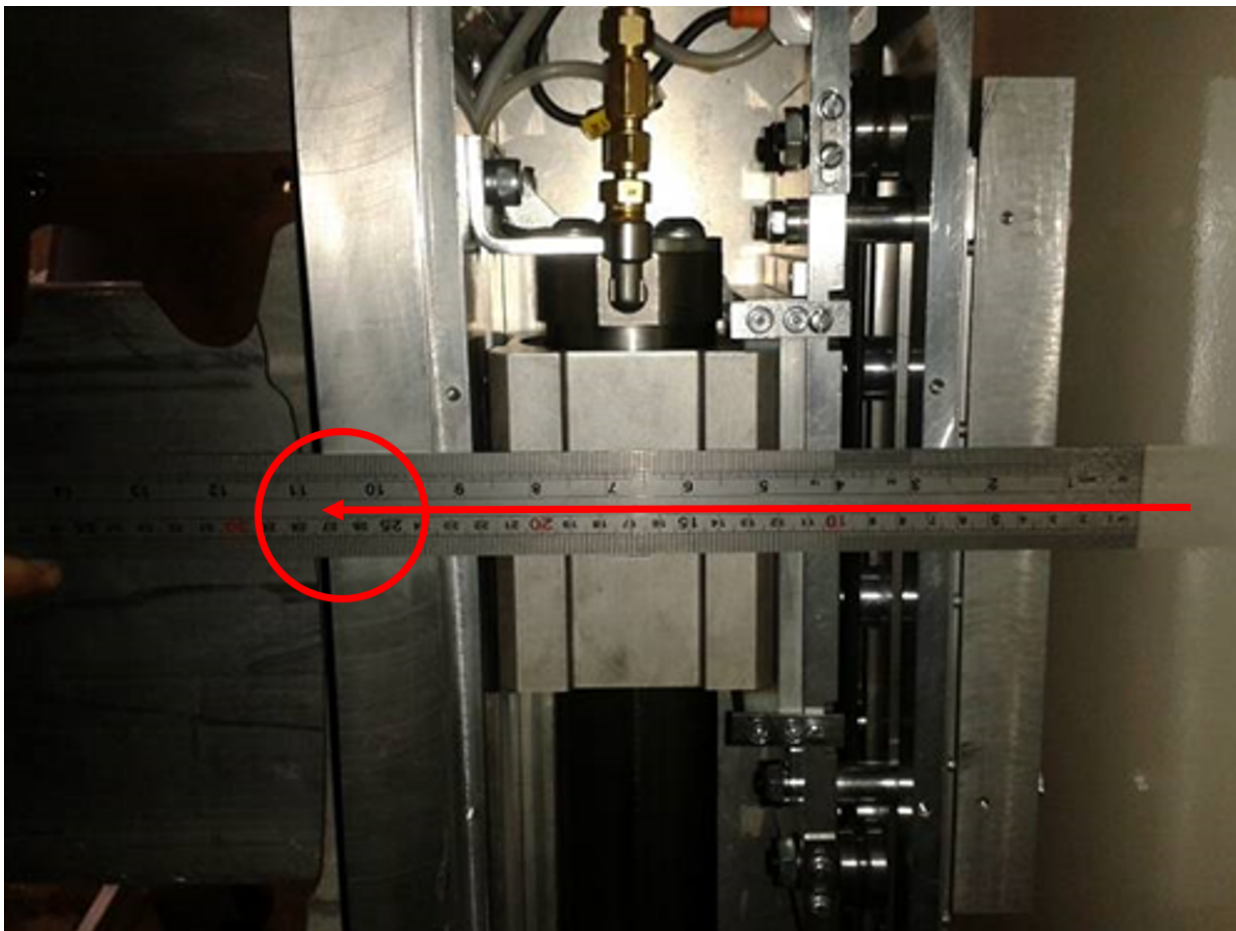


Figure 9.12-14 The setting distance for the shield to be centre around the process drum

4. Tighten up the bolts in the bearing housings to set this distance.

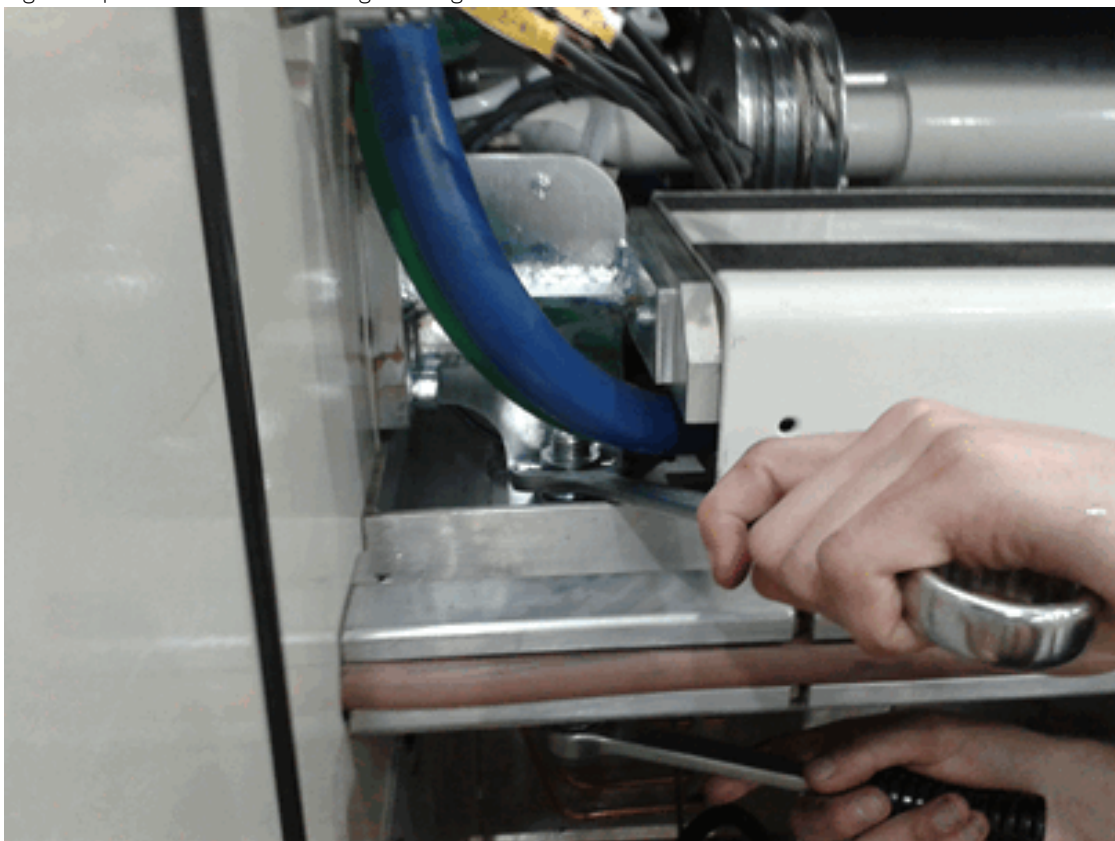


Figure 9.12-15 Example of a technician tightening the location bolts

5. Set the end stop adjustment bolts, at each of the shield to the maximum position as shown. There is one at each end of the shield.



Figure 9.12-16 A fully extended closing distance setting bolt

6. Raise the shield carefully ensuring no clashes with other equipment, particularly the drum or the 3 positioning sensors.



Figure 9.12-17 The coating shield lifted up into location.

7. Check the gap between the drum and the conductance seals using plastic shim. Gap should be 2-2.5mm along the length and also between the ends of the drum and the shield.

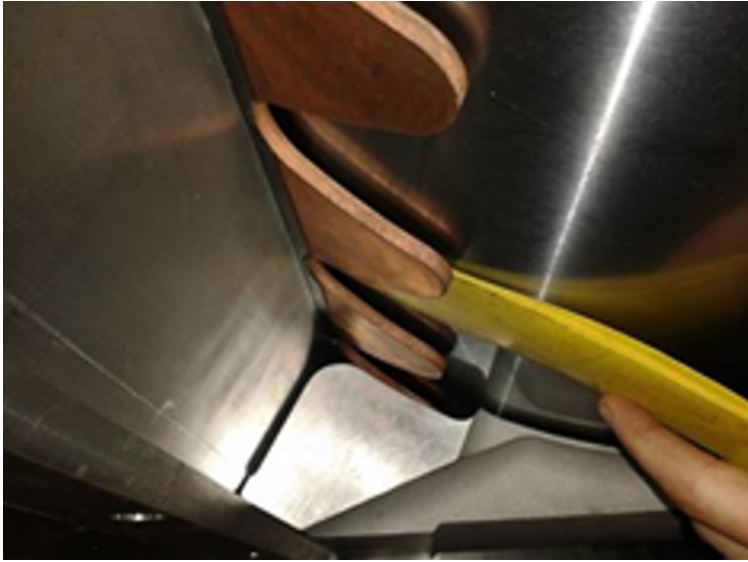


Figure 9.12-18 Example of a technician checking the gap around the drum.

8. To adjust the horizontal position of the shield using the plumber block jacking bolts (highlighted), after loosening the cap head bolts slightly.

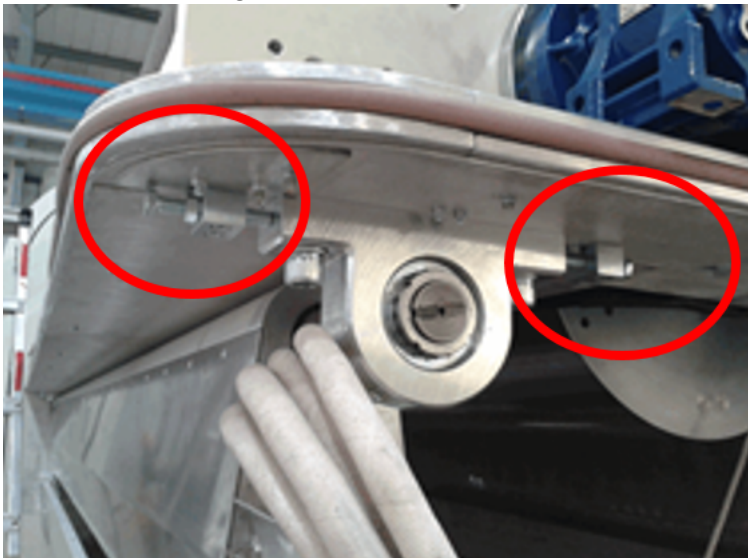


Figure 9.12-19 Jacking bolts

9. Adjust the height on the incoming side to the drum using the end stop adjustment bolts



Figure 9.12-20 Height adjustment screws.

10. When the gaps are correctly set at 2 to 2.5mm, ensure that all the fasteners are securely tightened.
11. Reconnect airlines and electrical cabling/leadthroughs.

Installing the Shield Support Safety Bolt

After raising the shield and checking the gap between the concave shields and the drum surface, the “safety shot bolt” has to be installed on the winding zone side of the partition.



Figure 9.12-21 Example of the assembly being placed in position

Ensure that the “BOLT” aligns with the opening in the bracket on the shield. Once located, install the locating bolts.

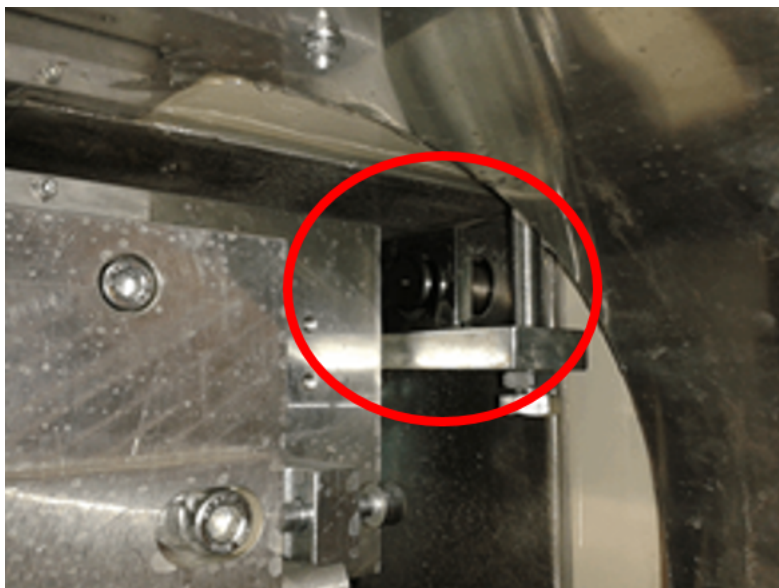


Figure 9.12-22 Example of the bolt placed through the opening in the shield.



Figure 9.12-23 The shield raised sensor.

Position Sensor

There are proximity sensors to allow monitoring of its position of the shield by the machine plc system.

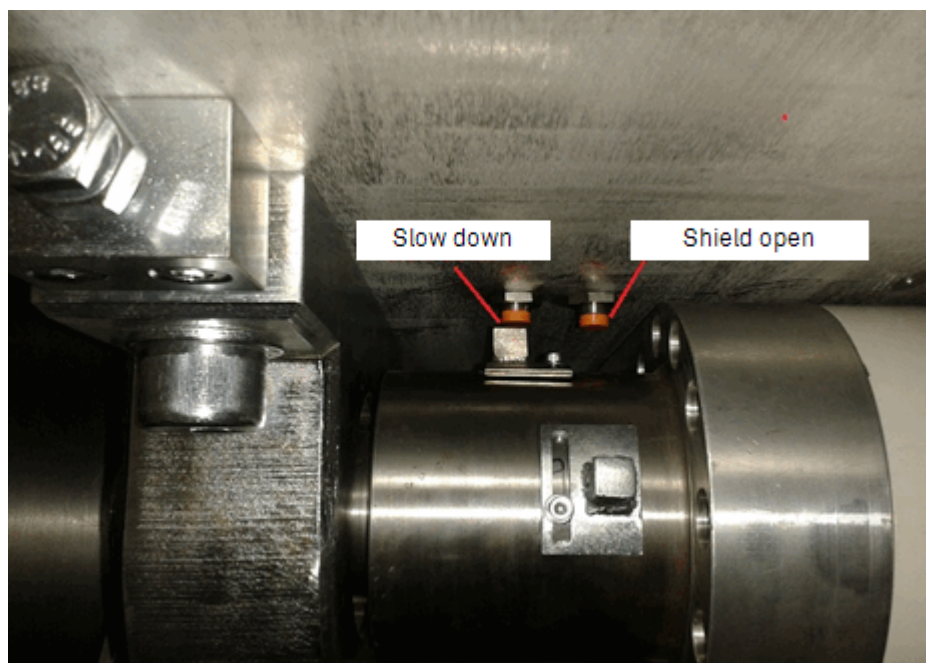


Figure 9.12-24 The coating shield proximity switches.

1. Set the 'slow down' sensor so the shield slows down with approximately 60mm to the fully raised position.
2. Set the 'open' sensor so the shield stops whilst there is still some tension in the cables and the shield is fully raised. Adjust the 'shield closed' sensor until it is positively reading the faceplate "end stop" bracket.



Figure 9.12-25 The shield full up (raised) sensor inside the evaporation zone.

9.13 Reporting Options/Data Logging

The computer can produce a number of reports of the machine concerning the operation, alarm status and production cycle times.

Reports Function

The REPORTS function screen allows the user to set up and generate a report on rolls of substrate previously coated and the running conditions of the machine. The screen is accessed from the REPORTS button at the top of the HMI screen.

Normally the screen below will be initially shown, this is the Select Screen.

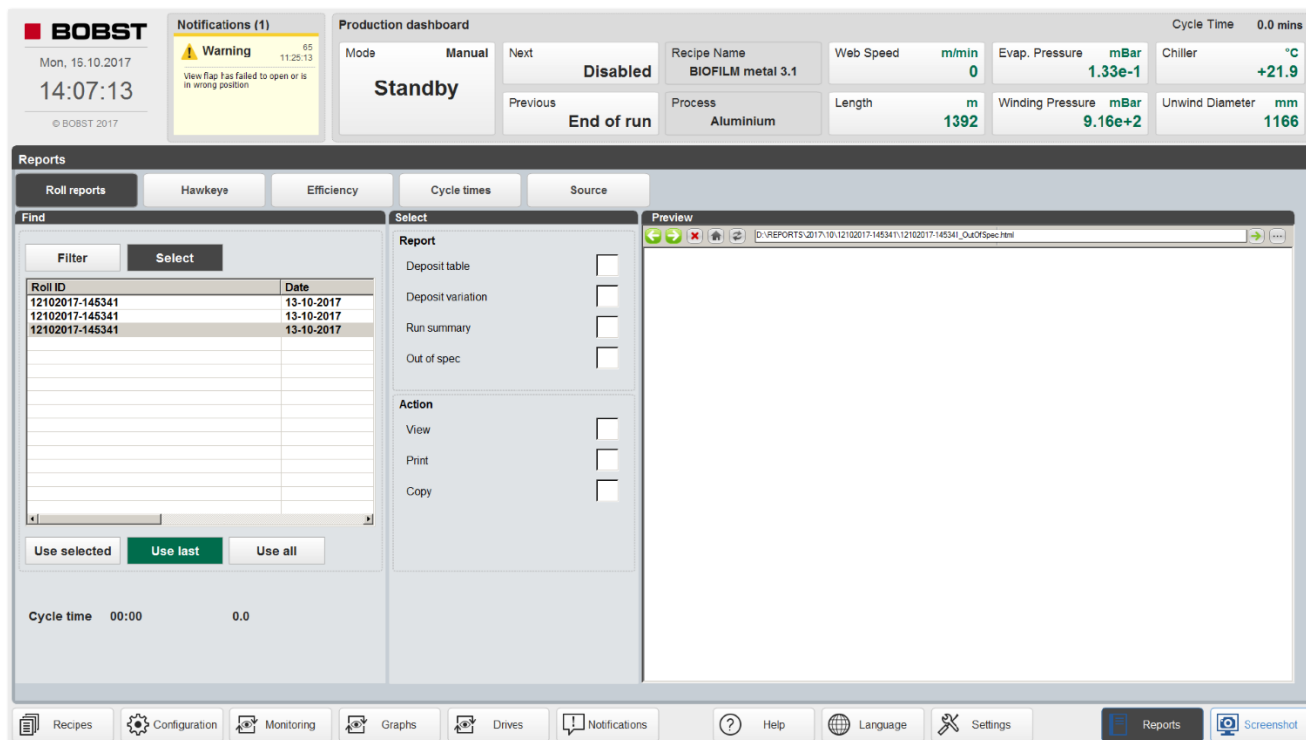


Figure 9.13-1 Reports Options Select Screen.

This screen allows the user the choice of the following types of report on rolls that have been processed to be printed or viewed in a preview mode.

If the tick is moved to the preview side, a preview screen of each report selected will be displayed on the HMI screen.

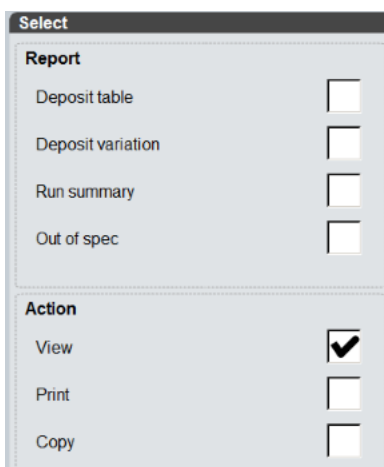


Figure 9.13-2 HMI Reports to Print Selection Screen.

The user can then choose to print this report to a previously set up printer on the machine.

Fields:	Explanation
Deposit Table	This report is called a “sampled report”. It shows the deposition readings throughout the coating cycle, on the probes selected for the width of the material being processed. This sampling is carried out on a time basis, and gives the time that the sample was taken and the length of the material processed. The default time span is every 30 seconds.
Deposit Variation	This report indicates the deposit measured in colour blocks to indicate if the deposition is in specification, or outside it.
Run Summary	This report shows the user the various process cycle settings and the amount of material processed for example.
Out of Specification	Whenever the deposition measured is outside the pre-set tolerance limits, the software will record the deposition level, the position across the width, the time and the position in the roll in length.

Table 9.13-1 Reports Table

	Note !
	The length measurement is from the start of the roll.

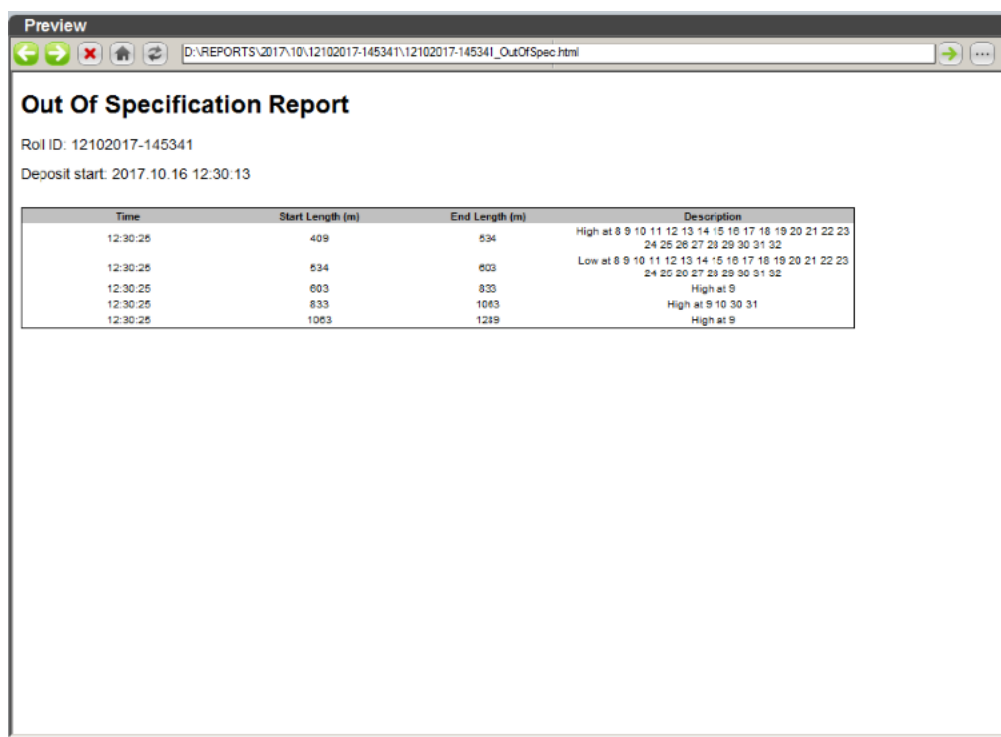


Figure 9.13-3 Typical preview of a report.

There are different functions at the top of the preview window to allow further actions.

Viewing Archived Reports

There is an archive function that allows the computer to store previous roll reports.



Figure 9.13-4 Setup Load Message.

This cross reference of the main data log achieves containing different data. This archive is currently limited to six months, and then the data is deleted.

As the SETUP function is used and the user uses the LOAD function, they will be asked if they want to allow the computer to designate an automated reference roll number to the cycle or not.

These unique numbers can be used later to select reports.

	Note !
	If the same Roll ID is used, then only one report will be written, that will be overwritten every cycle.

Select from data given in the setup screen, or type text into the field directly

Figure 9.13-5 Achieved Report Search parameters.

To find a previously achieved report, parameters can be set in the section of the screen shown above.

All reports are initially selected by choosing the selected fields.

Fields:	Explanation
Start Date	The start date for the search. Can be set quickly using the Today, This Week, or This Month buttons.
End date	The end date for the search. Can be set quickly using the Today, This Week, or This Month buttons.
This day	Select a specific day from the calendar
Shift	Allows the user to select a production shift, when the roll was processed
Operator	Allows the user to select a selection of rolls based upon the operator login name

Table 9.13-2 Search Fields

Once the search fields have been selected, the SELECT button should be used.

The results from the search will be shown on the screen.

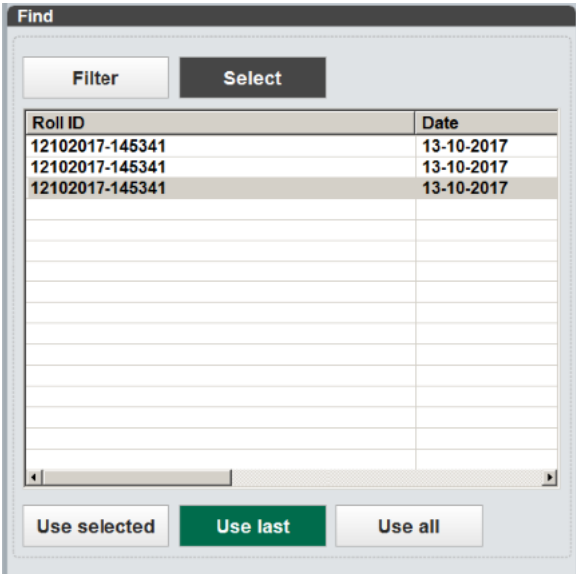


Figure 9.13-6 Report Search Results Table.

The details shown can be selected from the list content listing.

To select a file, it should be highlighted and then the USE SELECTED button be touched.

Alternatively, the last process run can be selected by touching the 'USE LAST' button.

Report Screen – Hawkeye

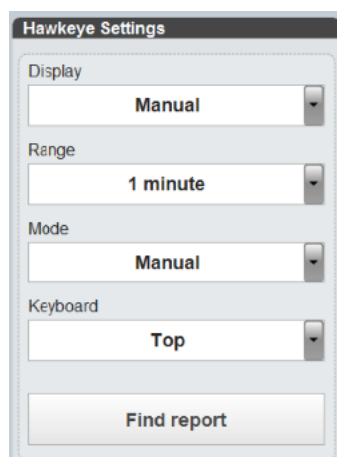
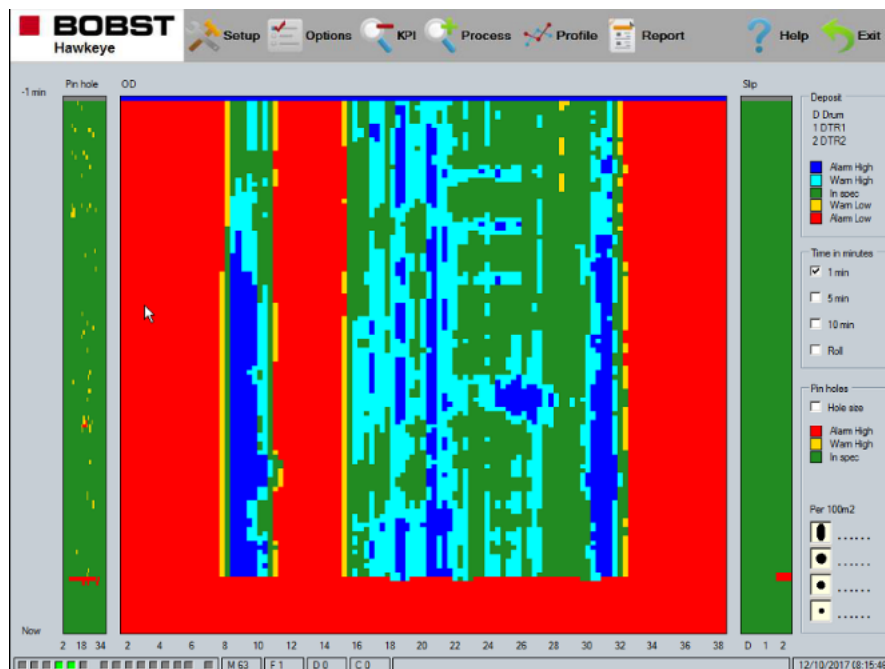
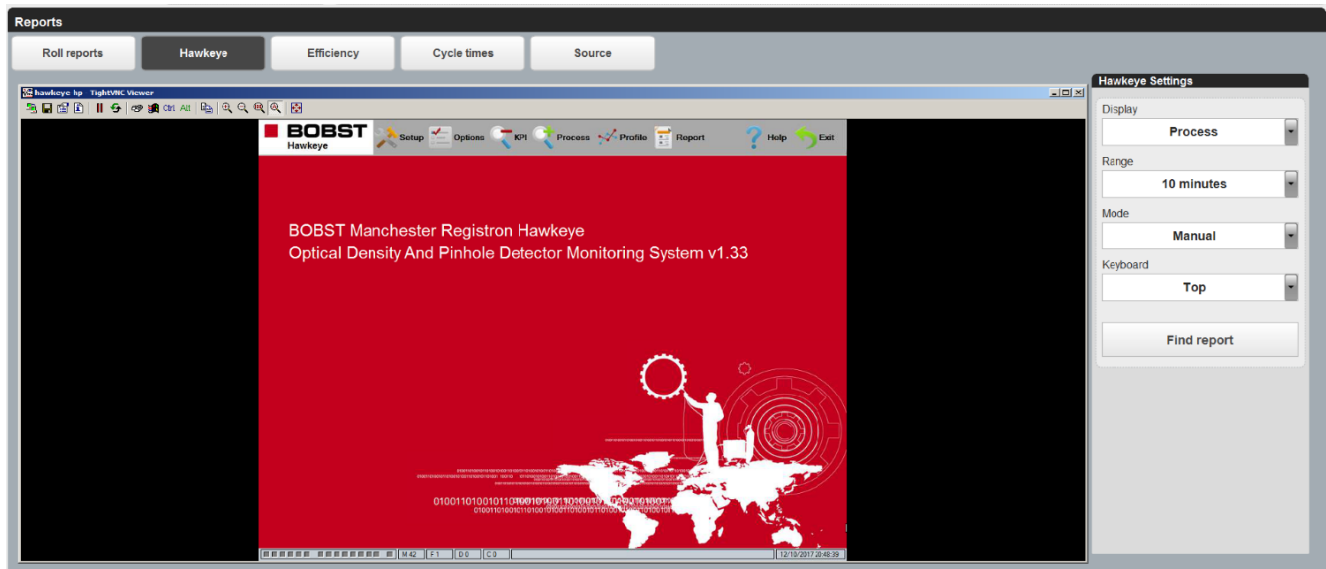


Figure 9.13-7 Hawkeye settings HMI screen.

Report Screen – Efficiency

This screen allows the user to quickly investigate how much of the substrate processed is actually within the pre-set specification.

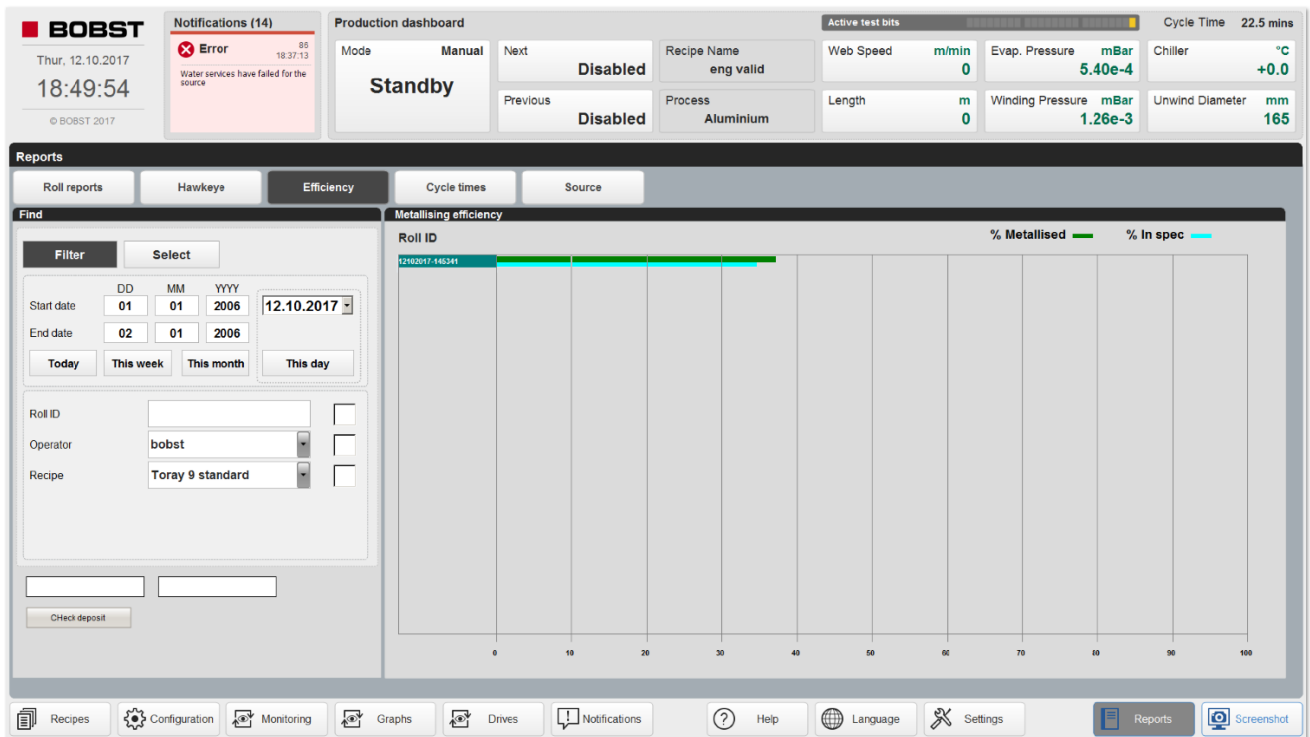


Figure 9.13-8 Machine Efficiency Report Screen.

These reports are shown against rolls/cycles. This is usually the last 28 cycles with the last cycle at the bottom of the screen.

The left hand column is colour coded to the shifts team using the machine.

Key:-

- Green is total metalizing cycle
- Light Blue is total metallized material within the specified limits as a percentage of the total material metallized

Report Screen – Cycle Times

This report allows the user to view the process cycles. The different periods in the overall production cycle are colour coded as per the guide at the top of the screen.

If the cycle time is long, the colour-coded chart may be extended. The red X on the right-hand side indicates this.

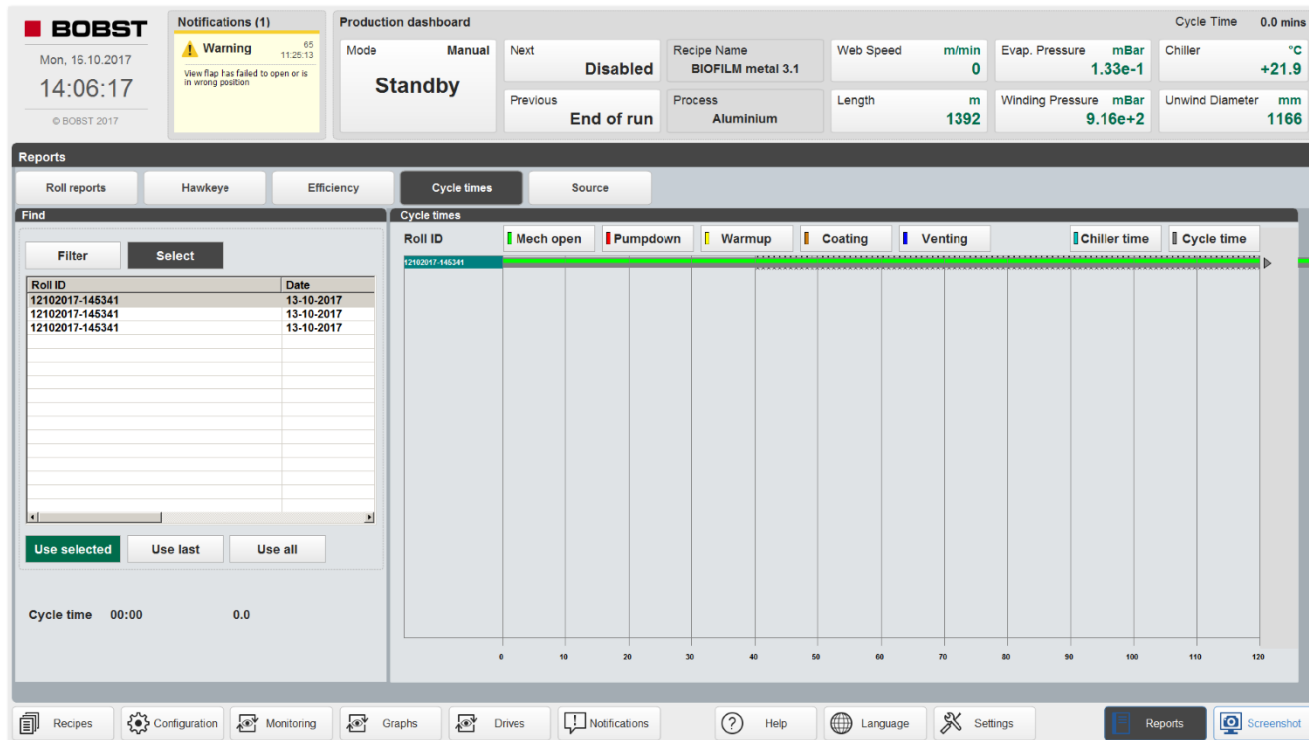


Figure 9.13-9 Machine Cycle Times Report Screen.

Pressing the index buttons at the top of the screen will set the display order with the selected item being first. The left hand column is colour coded to the shifts team using the machine.

Report Screen – Source

This HMI screen allows the user to monitor evaporator life cycle and the aluminium wire consumption.

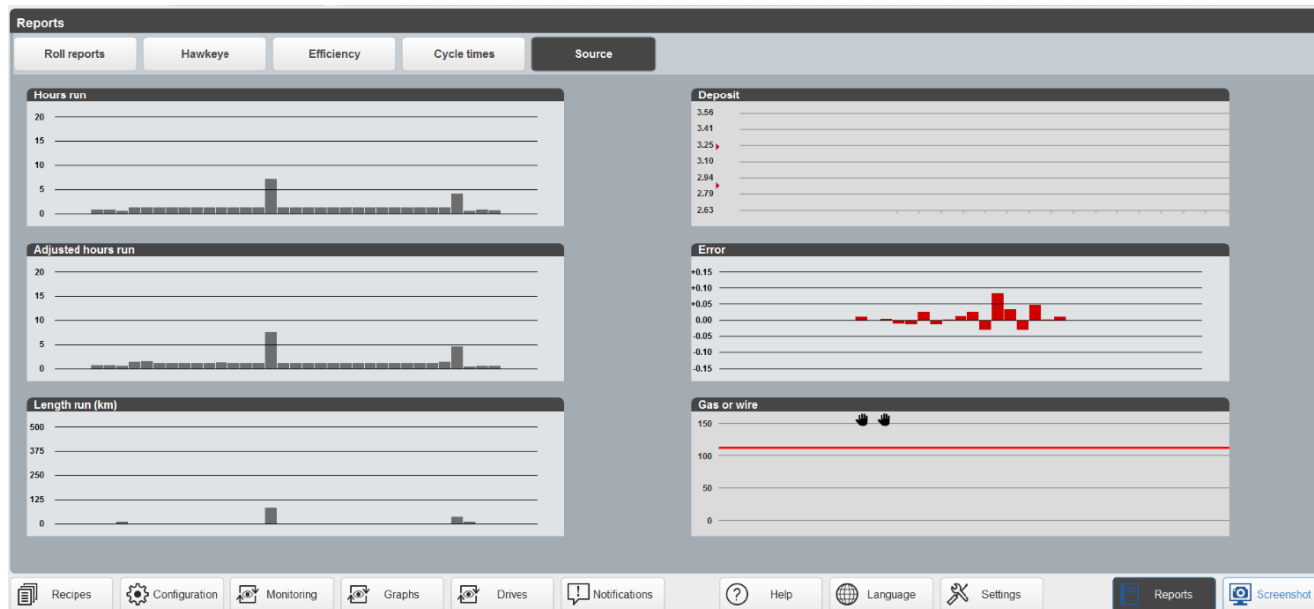
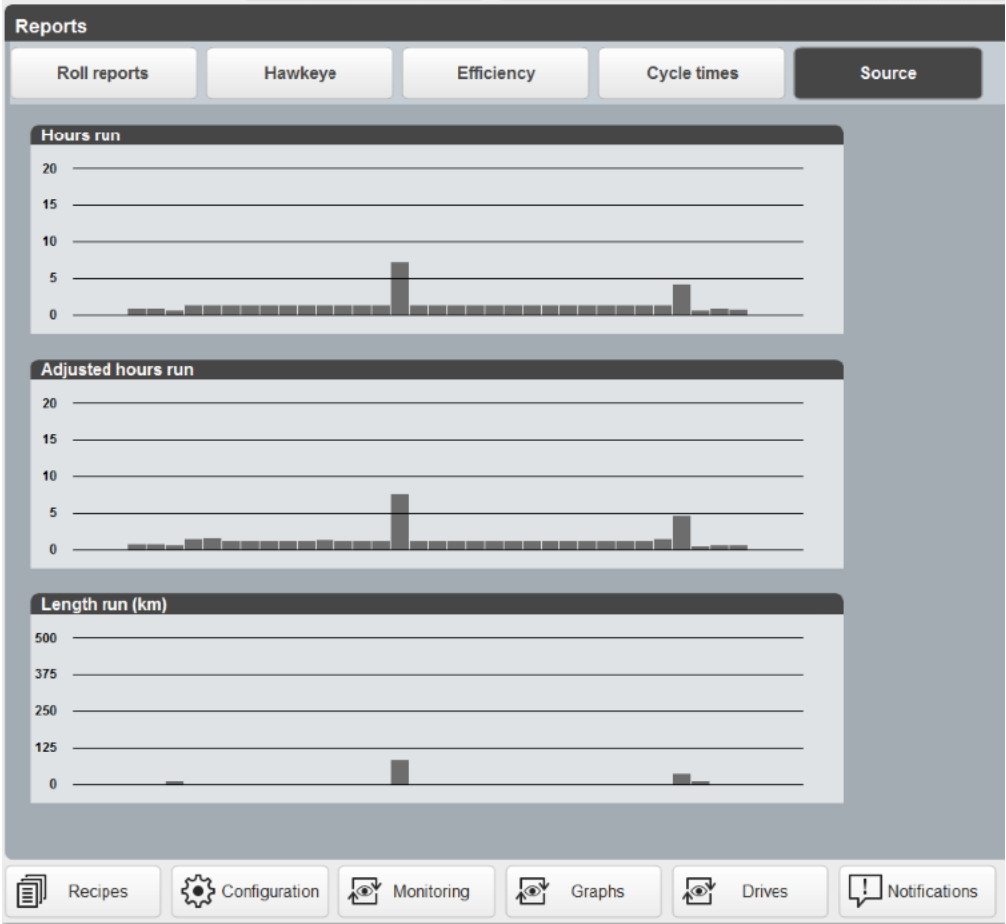
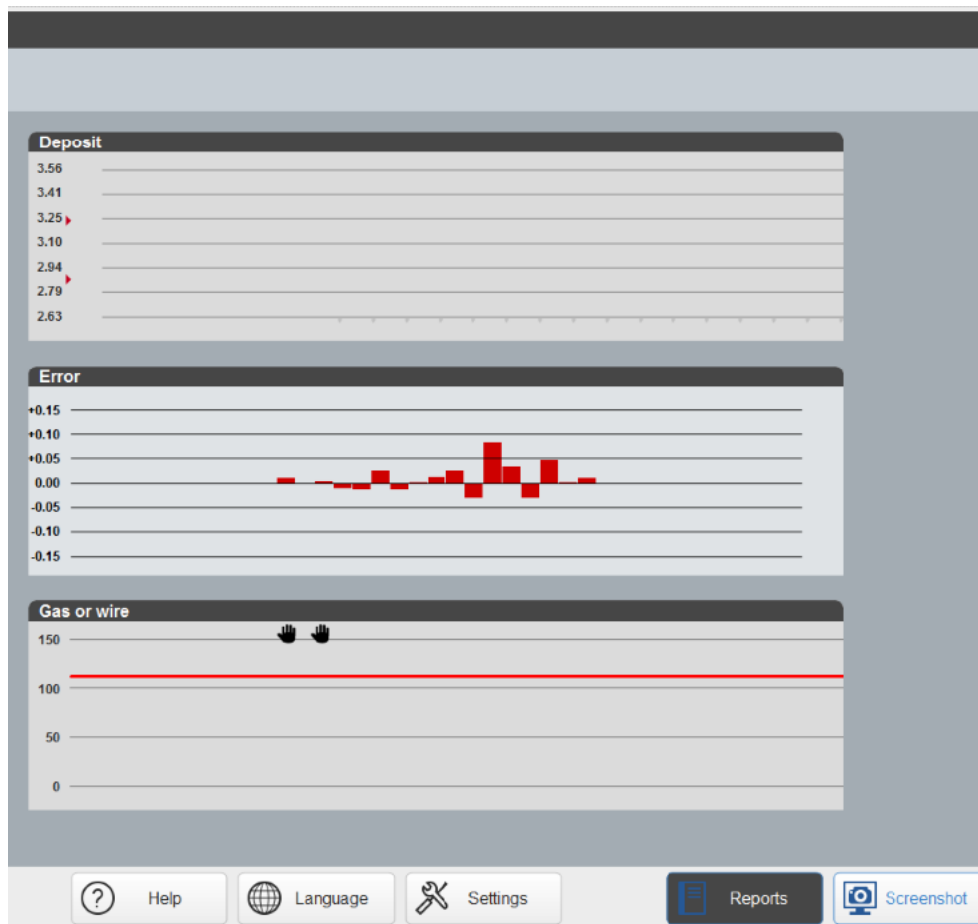


Figure 9.13-10 Source Report Screen.





The amount of wire of a reel will depend upon the size of the reel installed.

9.14 Maintenance HMI Screens

At the maintenance engineer log on level, there are a number of functions available to the user. Open the settings menu and then open the maintenance menu to view these functions.

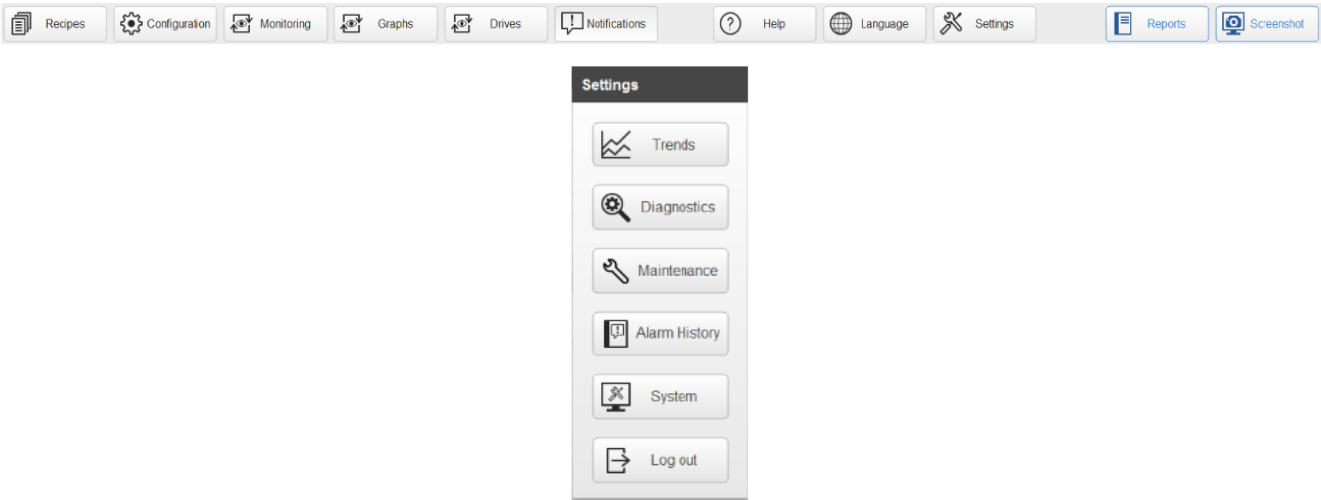


Figure 9.14-1 Navigation to the SERVICE HMI screen.

Maintain- Service Screen

This screen allows the maintenance engineer to change the time in hours that are currently used to make the user aware that there are maintenance tasks required to be carried out.

It also allows the maintenance engineer to reset the timers, to remove the warning from being displayed from the alarms screen, until the timer times out again.

	Note !
	The pre-set times are default values, based upon recommendations from the OEMs.

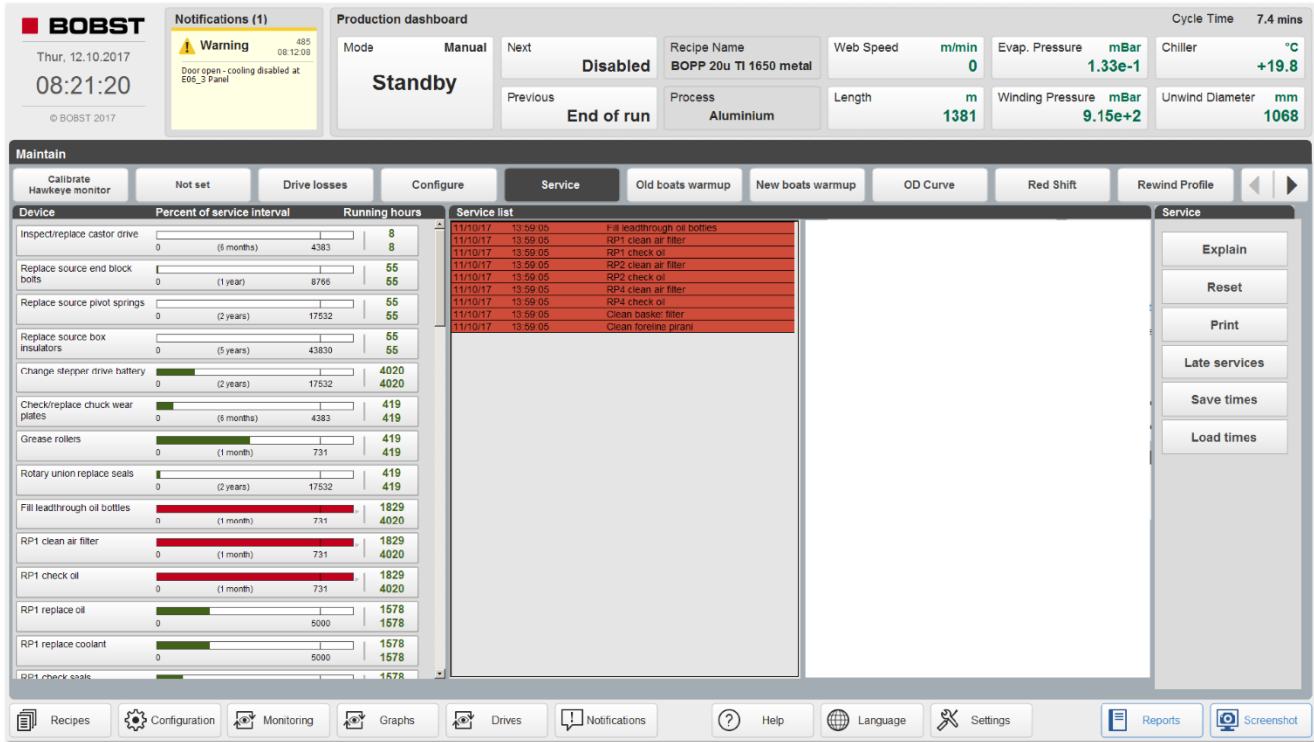


Figure 9.14-2 SERVICE HMI screen.

The device table displays a list of maintenance tasks:

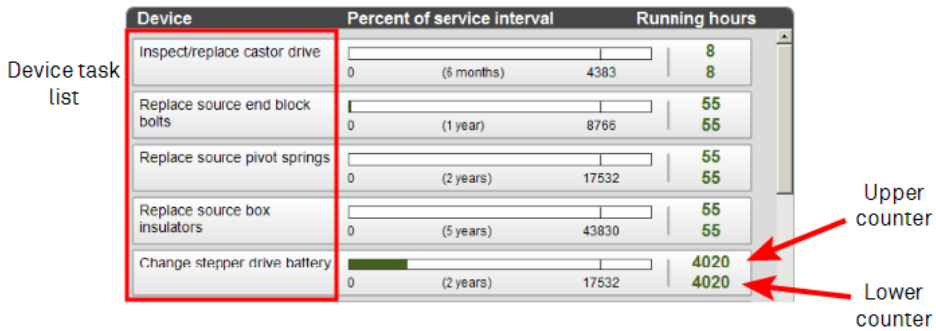


Figure 9.14-3 SERVICE running hours section of the HMI screen.

- Upper counter is the number of running hours between counter resets.
- Lower counter is continuous from when the machine was put into operation

The service list displays the outstanding service tasks:

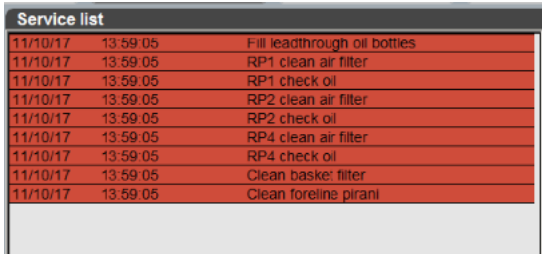


Figure 9.14-4 SERVICE HMI screen – service item list.

Pushbuttons are supplied for the tasks described below:

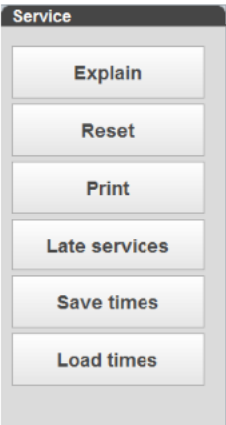


Figure 9.14-5 SERVICE HMI screen – function explanation.

Button	Explanation
Explain	Select an item in the device list to get an explanation of the servicing required.
Reset	Select an item in the device list to reset the run time.
Print	
Late service	
Save times	
Load times	

Table 9.14-1 Button function explanations

Maintenance Machine Screen

This screen has been provided to allow for some calibration functions, as described in other areas of the manual.

Figure 9.14-6 Maintenance machine HMI screen.

	Note !
	These values should not be changed unless the user fully understands the purpose and the outcomes of each of them.

Machine and Drives PLC Status Screens

These screens are provided to allow limited integration of the machine PLC CPU modules. The screens are standard to Siemens products.

Figure 9.14-7 Siemens 315F CPU interface screen.

Figure 9.14-8 Siemens 317-2 CPU interface screen.

Brookes Automation Cryogenerator Interface Screen

This screen has been provided by the OEM and the relevant details on the screens, parameters and values are described in the OEM’s manual.

Figure 9.14-9 Brookes Automation cryogenerator interface screen.

9.15 Rear HMI Interface Unit

The rear HMI Interface Unit allows the user to take control of the operation of the substrate winding drive system. It is placed at the rear so that the operator can see the devices being operated, and cannot be operated remotely for safety reasons.



Figure 9.15-1 Rear local control HMI station.

Main Screen

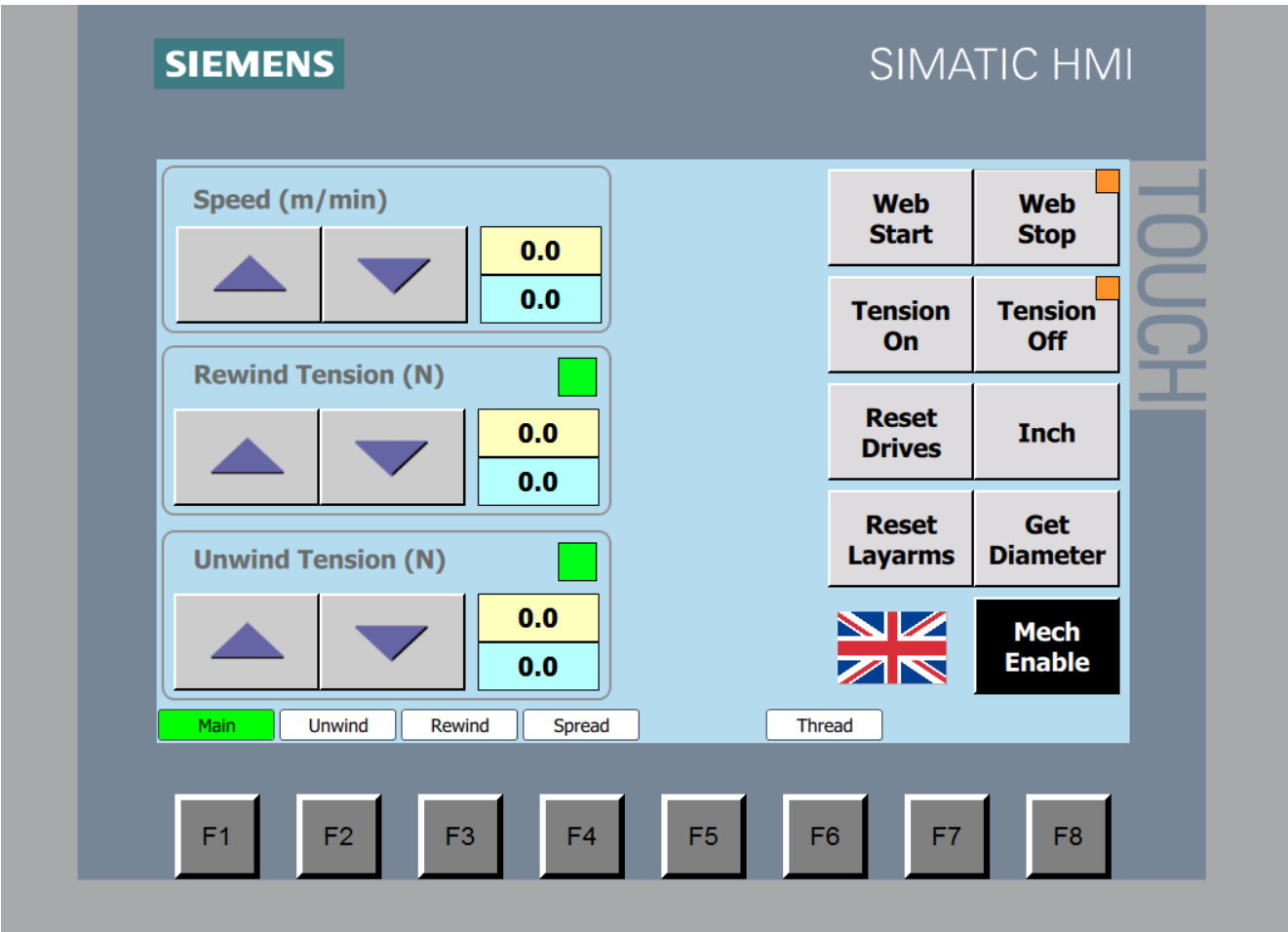


Figure 9.15-2 Main interface screen.

The following buttons are shown on every screen. An orange square appears in the upper right-hand side of a button when it is enabled.

Button	Explanation
Mech Enable	Places the machine control in an enabled state if the required conditions are met. The operator should walk through the machine before closing to ensure no personnel are inside. The operator has 30 seconds to close the chamber. This button is used in the control logic for the movement of the winding cart into the chamber.
Web Start	Starts the web.
Web Stop	Stops the web.
Tension On	Applies tension to the substrate if the required conditions are met.
Tension Off	Removes tension from the substrate.
Reset Drives	Reset the drive system to a healthy condition following a fault condition.
Inch	Inch (jog) the substrate through the winding mechanism.
Reset Layarms	Reset the drive system layarms to their home positions if the required conditions are met.
Get Diameter	Moves the layarms into the rolls to measure their diameter for the drive system to operate correctly.

Table 9.15-1 Explanation of main screen functions

The following buttons only appear on this screen. In all cases, the set point is number in the blue box, with position feedback in the yellow box. Use the arrow buttons to set the value.

Function	Explanation
Speed M/min	Sets the required line speed.
Rewind Tension (N)	Sets the tension set point.
Unwind Tension (N)	Sets the tension set point.

Table 9.15-2 Explanation of main screen functions

Unwind Interface Screen

Figure 9.15-3 Unwind Layarm screen.

Function	Explanation
Unwind Layarm	Sets the rewind layarm gap in mm or contact pressure in newtons (N), using the arrow buttons. The set point is number in the blue box, with position feedback in the yellow box.
Unwind Layarm Manual Buttons	Moves the layarm assembly IN towards the core/roll or OUT away from the roll

Table 9.15-3 Explanation of screen functions

Function	Explanation
Unwind Sidelay	Moves the unwind roll left and right once it has been chucked. The green indicator will go red when full movement is reached.

Table 9.15-3 Explanation of screen functions

Rewind Interface Screen

Figure 9.15-4 Rewind Layarm screen.

Function	Explanation
Rewind Layarm	Use these buttons to remotely set the rewind layarm gap in mm, or contact pressure in newtons (N), using the arrow buttons. The set point is number in the blue box, with position feedback in the yellow box.
Rewind Layarm Manual Buttons	Use these buttons to move the layarm assembly IN towards the core/roll or OUT away from the roll
Rewind Sidelay	Use these buttons to move the rewind roll left and right, once it has been chucked. The green indicator will go red when full movement is reached.

Table 9.15-4 Explanation screen functions

Spreaders Interface Screen

Figure 9.15-5 Spreaders Screen.

Function	Explanation
Drum Spreader angle	This function allows the user to remotely move the amount of spreader apex in degrees using the arrow buttons. The set point is number in the blue box, with position feedback in the yellow box.
Rewind Spreader Angle	This function allows the user to remotely move the amount of spreader apex in degrees, using the arrow buttons. The set point is number in the blue box, with position feedback in the yellow box.

Table 9.15-5 Explanation of screen functions

Threader Interface Screen

Figure 9.15-6 Threader Screen.

Function	Explanation
Web threader Interlocks	All these interlocks have to be green, to allow the threader to be utilised.
Run	Touching the run button, will start the movement of the web threader.

Table 9.15-6 Explanation of screen functions

Bonfanti Crane Interface Screen

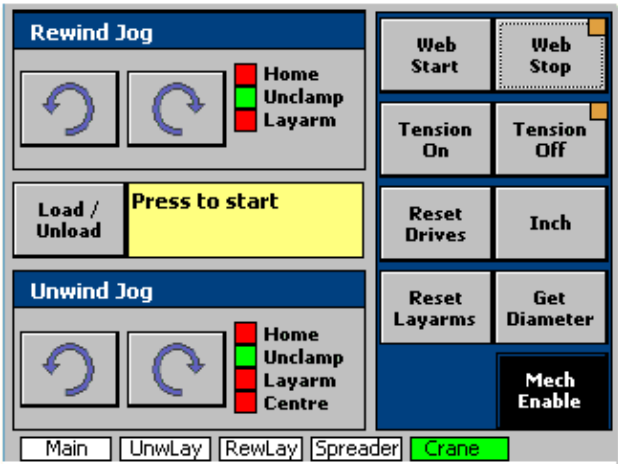


Figure 9.15-7 Crane Interface Screen.

Adjustable Parameter/Button	Description
Counter clockwise/clockwise icon	There is a manual function to allow the driven chuck to be rotated, by the operator slowly either clockwise or counter-clockwise. The “home indicator will change colour from red to green, when the chuck is in the correct position.
Load/Unload Button	Operator steps the load/unload sequence using this button
Home	Chuck is in the correct position to allow removal or placement of a roll/shaft
Unclamp	Indicates that the chuck is mechanically unclamped by the operator
Centre	Indicates that on machines with sidelay function, the chuck has been centred in the machine.
Layarm	Indicates that the layarm has been positioned at its home position.

Table 9.15-7 Crane HMI screen functional descriptions

E- Stop Push-button:

Enabling this push-button will place all rotating components into a power down condition.

9.16 Control Interface Unit Pushbuttons

The following section describes the purpose and location of all the control push-buttons and switches on the machine.

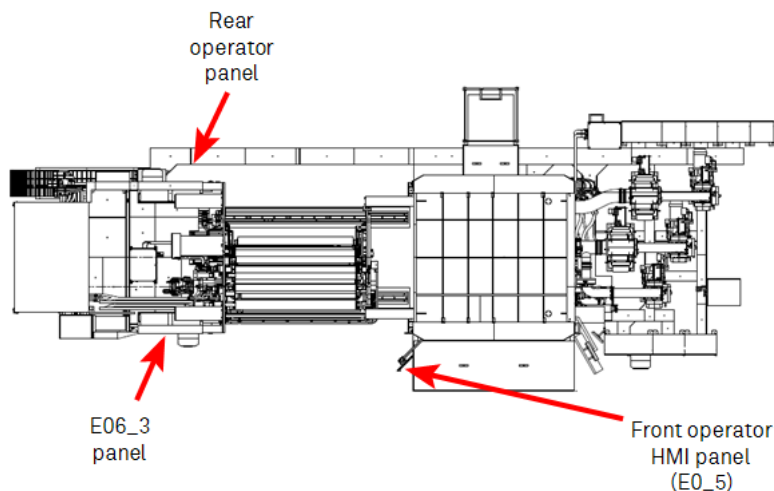


Figure 9.16-1 Location of control stations.

Front Operator HMI Panel



Figure 9.16-2 The HMI interface screen.

Pushbutton	Description
E-stop	Pressing this button will place the machine in an emergency stop condition. This is described elsewhere in this manual
Enable	This push-button is used in conjunction with the winding cart OPEN OR CLOSED push buttons. It is also required to be pressed with the VACUUM button
E-stop reset	When the Emergency stop circuit has been utilised this button is used to reset the control circuit back into an operational state. This action is also required when starting the machine, from a powered down condition.
G-stop reset	When the G-stop circuit has been, utilised, this button is used to reset the control circuit back into an operational state. This action is also required when starting the machine from a powered down condition
View source	This pushbutton is used to allow the viewing of the evaporation source evaporators.
Shield up	This push-button is used to raise the coating shield when the chamber is fully open.

Table 9.16-1 List of pushbuttons and function descriptions

Pushbutton	Description
Shield down	This push-button is used to lower the coating shield when the chamber is fully open.
Shutter open	This pushbutton is provided to allow the user to open the coating shutter if the mode of operation has been selected to “manual” control.
Shutter close	This pushbutton is provided to allow the user to close the coating shutter if the mode of operation has been selected to “manual” control.
Inch	This pushbutton is provided to allow the user to rotate one or more rollers in the winding mechanism, depending upon the “tension condition” of the substrate.
Open	This push-button is used to move the winding cart out of the chamber to its fully open position. Use in conjunction with the ENABLE push-button.
Vacuum	This push-button is used to start the vacuum sequence in conjunction with the ENABLE push-button. Pressing both buttons will also drive in the mechanism cart for a time limited period.
Close	This push-button is used to drive the winding cart into the chamber from its parked open position. Use in conjunction with the ENABLE push-button.

Table 9.16-1 List of pushbuttons and function descriptions

E06_0 Panel

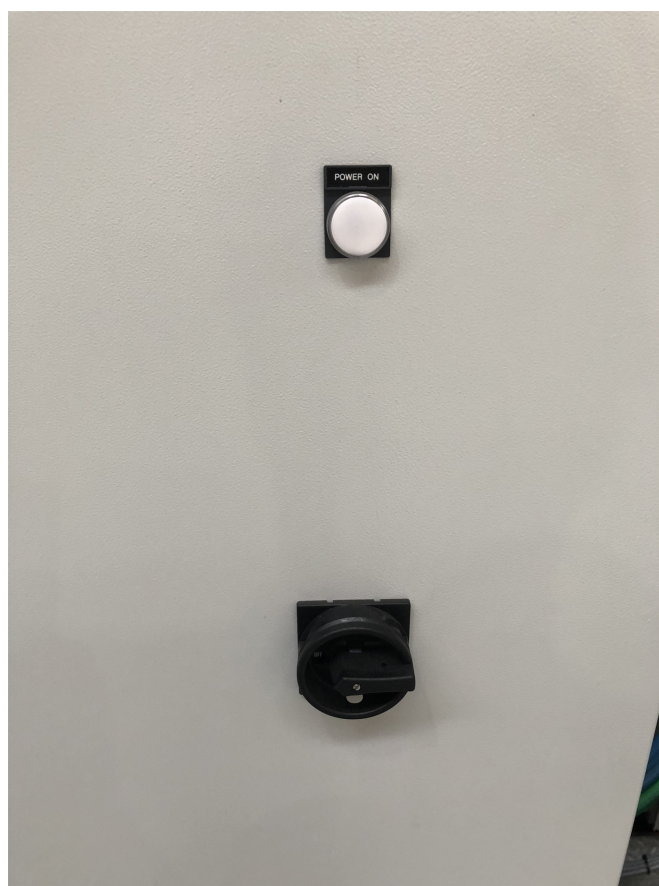


Figure 9.16-3 E06_0 electrical panel controls.

Pushbutton	Description
Lockout Isolator	This lockout isolator is provided to ensure that the machine is in a safe condition when it has been switched to the OFF position.

Table 9.16-2 List of pushbuttons and function descriptions



Warning !

Under no circumstances must any person enter the chamber unless they have locked out the isolator.

Rear Operator Panel



Figure 9.16-4 The rear operator panel.

Pushbutton	Description
E-stop	Pressing this button will place the machine in an emergency stop condition. This is described elsewhere in this manual

Table 9.16-3 List of pushbuttons and function descriptions

SECTION 10.0 PROCESS CONSUMABLES

The process consumable items required for the machine are, evaporators, aluminium wire, lubricating oils, vacuum greases, graphite tape and release paint.

There are some mechanical parts upon the machine, which will require changing periodically, and these are classed as wear parts. These include for example; bearings, bushes (bushings), boat contact posts, and wire feed tubes

Electrical items such as internal fluorescent light tubes are classed as consumable items, as they are expected to "burn out" over a period of time.

This section describes some of the process consumables for the K5 Expert in the following chapters:

- [10.1 Evaporated Aluminium Release Solution on page 505](#)
- [10.2 Graphite Tape on page 507](#)
- [10.3 Ceramic Evaporators on page 508](#)
- [10.4 Aluminium Wire on page 509](#)
- [10.5 Lubricants on page 510](#)
- [10.6 Subsystem Antifreeze on page 511](#)
- [10.7 Evaporation Section Contact Blocks and Wire Feeder tubes on page 512](#)
- [10.8 Process Gases on page 514](#)